Piter Z. Garcia Bautista

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Research Summary

NSF Robert Noyce Scholar and graduate researcher working at the intersection of equitable machine learning, reproducible experimentation, and high-stakes decision support. Current work spans fairness-aware learning, contextual and sequential decision workflows, and robust evaluation under shifting operational conditions. For maritime environmental monitoring, I am particularly interested in adaptive AI methods that remain reliable under noisy sensor streams, constrained communication, and non-stationary dynamics.

Education

Rochester Institute of Technology — Expected Aug 2026
Master of Science, Data Science — GPA 3.9 — *Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644), Applied Data Science I (DSCI-601), Bioinformatics Algorithms (BIOL-630), Data-Driven Knowledge Discovery (ISTE-780).

University of Rochester, Warner School of Education — Expected Aug 2026
Master of Science track, Teaching Computer Science K–12 — GPA 4.0 — *Rochester, NY*

- NSF Robert Noyce Teacher Scholarship Program ($30,000 full funding).
- Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology — 2015
Master of Science, Computer Science — GPA 3.2 — *Rochester, NY*

- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics.

Farmingdale State College — 2012
Bachelor of Science, Computer Engineering Technology — GPA 3.3 — *Farmingdale, NY*

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit, Dual BS Scholarship.

Current Research Experience & Projects

RIT & University of Rochester — 2024 – Present
Lead Researcher, EQUITAS Framework for Diagnostic Justice — *Rochester, NY*

- Developed an interdisciplinary framework integrating fairness-aware ML, bioinformatics, and trauma-informed analysis.
- Built reproducible bias-assessment workflows and reported equity gaps across algorithmic variants.

- Coordinated cross-domain collaboration across computing, education, and health research contexts.

Rochester Institute of Technology 2024 – Present
Principal Investigator, Equitable Bioinformatics Project (ISTE-780) *Rochester, NY*

- Applied systematic fairness analysis to RNA secondary structure prediction pipelines.
- Implemented reproducible evaluation codebase with automated metric reporting and statistical testing.

Rochester Institute of Technology Fall 2025 – Present
Graduate Assistant, Quantum and AI Research *Rochester, NY*

- Run controlled experiments for contextual bandit and routing workflows under stochastic and adversarial settings.
- Build traceable experiment infrastructure across local, Colab, and GCP environments for robust comparison.

Teaching & Mentoring Experience

Varsity Tutors & Independent Practice 2021 – Present
STEM Tutor and Academic Mentor *Remote & In-Person*

- Tutor Computer Science, Data Science, Mathematics, and Statistics for diverse learners.
- Design culturally responsive instruction and mentoring support for underrepresented students in STEM.

Rochester Institute of Technology 2013 – 2015
Faculty Assistant, Graduate Office *Rochester, NY*

- Supported graduate program operations, student documentation, and faculty coordination.

Selected Publications & Academic Contributions

- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025).
- *Designing Equitable AI Systems for Diagnostics: An IDAI700 Research Portfolio* (Project Report, 2025) — ethical AI framework addressing diagnostic justice for providers and patients from marginalized communities.
- *Trauma-Informed Bioinformatics for Diagnostic Justice: An EQUITAS Framework* (ED-447 Capstone, 2025).
- *RNA Secondary Structure Prediction and Visualization: Implementing the Nussinov Algorithm with MFE Enhancements* (BIO-630 Final Project, 2025).
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Report, 2025) — bandit learning with equity constraints for high-stakes sequential decision-making in adaptive systems.
- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression for clinical risk assessment under distribution shift.
- *ASL Recognition Application: Image Processing for Sign Language Translation* (MS Thesis, 2015).

Technical Skills

Programming: Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, fairness-aware ML, contextual learning workflows
Data & Analysis: pandas, NumPy, SciPy, SHAP analysis, statistical testing, reproducible benchmarking
Infrastructure: AWS, GCP, Docker, Git/GitHub, Linux, LaTeX

Professional Experience

VEDADATA Inc. Sep 2022 – Aug 2024
Data Solutions Engineer *New York, NY*

- Designed and implemented data pipelines and analytics workflows using Python and cloud services.

VIOME Inc. Jan 2021 – Sep 2022
AI Data Solutions Engineer *New York, NY*

- Built ML workflows for healthcare applications with production deployment and monitoring support.

Selected Fit for NTNU AI-Enabled Maritime Systems Position

- Strong fit for adaptive maritime monitoring research through experience with dynamic decision workflows, reliability analysis, and constrained-system evaluation.
- Demonstrated reproducibility discipline for multi-run experimental design, statistical validation, and transparent reporting.
- Interdisciplinary AI background spanning method development, applied data systems, and high-stakes deployment contexts.